

The ngspice **optimize** command

A friendly user manual for the built-in parameter optimizer

Ngspice + OpenVAF Enhancements project

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The ngspice **optimize** command — a friendly user manual

A built-in parameter optimizer for ngspice (Enhancement-130, with least-squares curve fitting added in Enhancement-143).

This guide explains how to use the **optimize** command from scratch. You do not need any background in optimization or numerical methods — if you can write a small SPICE netlist and run it, you can use the optimizer. We build up from the simplest possible circuit and end with a few genuinely useful design tasks, with pictures at every step.

1. What problem does it solve?

When you design a circuit you usually know what you want the circuit to do — a gain of exactly 0.5, a filter that rolls off at 1 kHz, an output that reaches 0.9 V after 1 ms — but you don't know the exact component values that achieve it. The normal way to find them is trial and error: change a resistor, run the simulation, look at the result, change it again, and repeat until it's close enough.

The **optimize** command does that loop for you, automatically and quickly. You tell it:

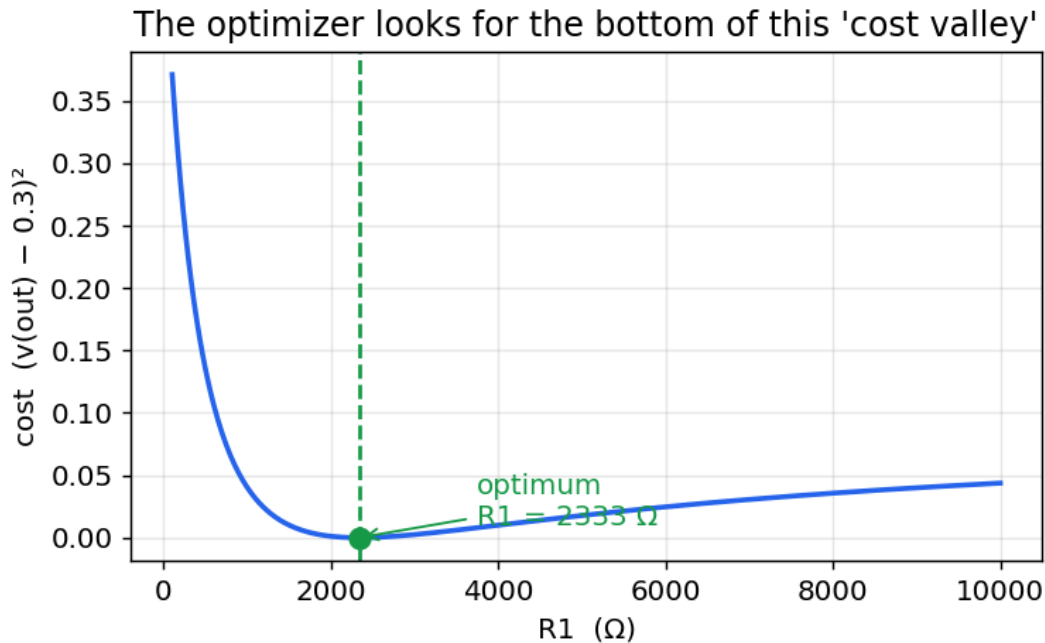
- which knobs to turn (which component values it may change, and their allowed range),
- what to run (a normal ngspice analysis: `op`, `ac` ..., `tran` ...),
- what "good" means (a small mathematical expression that is *zero* when the circuit does exactly what you want),

and it turns the knobs for you until the circuit is as close to your goal as possible.

The mental picture

Think of the thing you are trying to make small — how far the circuit is from your goal — as the height of a landscape. Turning a knob moves you east–west; the "badness" of the result is how high up you are. Your goal sits at the very bottom of a valley. The optimizer is like a hiker who can't see the whole map but can feel which way is downhill, and keeps stepping down until they reach the bottom.

Here is that landscape for our very first example (turning one resistor R1, trying to make the output 0.3 V). The optimizer's job is simply to find the bottom of this valley:



The number plotted vertically is called the cost (or *objective*): it is 0 when the circuit is perfect and grows as the circuit gets worse. You define the cost; the optimizer only ever tries to make it small.

2. The command at a glance

You run `optimize` inside a `.control ... .endc` block, after your circuit is loaded:

```
optimize -param <name> <init> <lo> <hi>    [-param ...]
        -analysis <command ...>
        ( -minimize <expression ...>          (one goal)
          | -target <expr> <value> [<weight>] [-target ...] ) (fit several)
        [-method nm|lm] [-maxiter <N>] [-tol <T>] [-verbose]
```

Part	Meaning
-param name init lo hi	A knob to turn. name is a device (like R1, C1) or a device parameter. init is where to start, lo/hi are the smallest/largest values allowed. Repeat -param for each knob (up to 16).
-analysis <cmd>	The simulation to run every time it turns the knobs — an ordinary ngspice command such as <code>op</code> , <code>ac dec 20 1 1meg</code> , or <code>tran 1u 1m</code> . Give several to combine analyses in one fit (see §8).
-minimize <expr>	The cost, for a single goal. Any ngspice expression over the results that should be zero when the circuit is perfect. A very common shape is <code>(something - target)^2</code> .

Part	Meaning
<code>-target <expr> <val> [<w>]</code>	A measurement to fit (§8). Repeat to fit many at once; the optimizer minimizes the sum of squared residuals $w \cdot (\text{expr} - \text{val})$. Use <code>-target</code> or <code>-minimize</code> , not both.
<code>-method nm lm</code>	(optional) force Nelder-Mead (nm) or Levenberg-Marquardt (lm). Default: a <code>-target</code> fit uses lm, a <code>-minimize</code> goal uses nm.
<code>-maxiter N</code>	(optional) stop after at most N steps. Default 100.
<code>-tol T</code>	(optional) stop when the cost stops improving by more than T. Default 1e-6.
<code>-verbose</code>	(optional) print the cost after every step so you can watch it fall.

A couple of friendly details so you don't trip up:

- `-analysis` and `-minimize` gobble up all the words after them until the next `-word` flag, so you can write `-analysis ac dec 20 1 1meg` or `-minimize (v(out)-0.3)^2` without quotes.
- A negative lower bound like `-5` is understood as a number, not a flag (flags start with `-` followed by a *letter*).
- The hundreds of little simulations the optimizer runs are silent by default, so your screen isn't flooded. Add `-verbose` if you want to watch progress.

3. Example 1 — the simplest possible case (one knob)

A voltage divider: a 1 V source drives two resistors in series, and we measure the voltage at their mid-point, out. Basic circuit theory says $v(\text{out}) = R2 / (R1 + R2)$. With $R2 = 1 \text{ k}\Omega$, if we want $v(\text{out}) = 0.3 \text{ V}$, the exact answer is $R1 = 2333.3 \text{ }\Omega$. Let's pretend we don't know that and let the optimizer find it.

Voltage divider: tune R1 so $v(\text{out}) = 0.3 \text{ V}$

```
V1 in 0 dc 1
R1 in out 1k
R2 out 0 1k
```

```
.control
optimize -param R1 1k 100 10k -analysis op -minimize (v(out)-0.3)^2
print v(out)
.endc
.end
```

Reading the optimize line in plain English: *"turn R1 (start at 1 k Ω , keep it between 100 Ω and 10 k Ω); each time, run an operating-point (op) analysis; and try to make $(v(\text{out}) - 0.3)^2$ as small as possible."* That expression is zero exactly when $v(\text{out})$ equals 0.3, which is what we want.

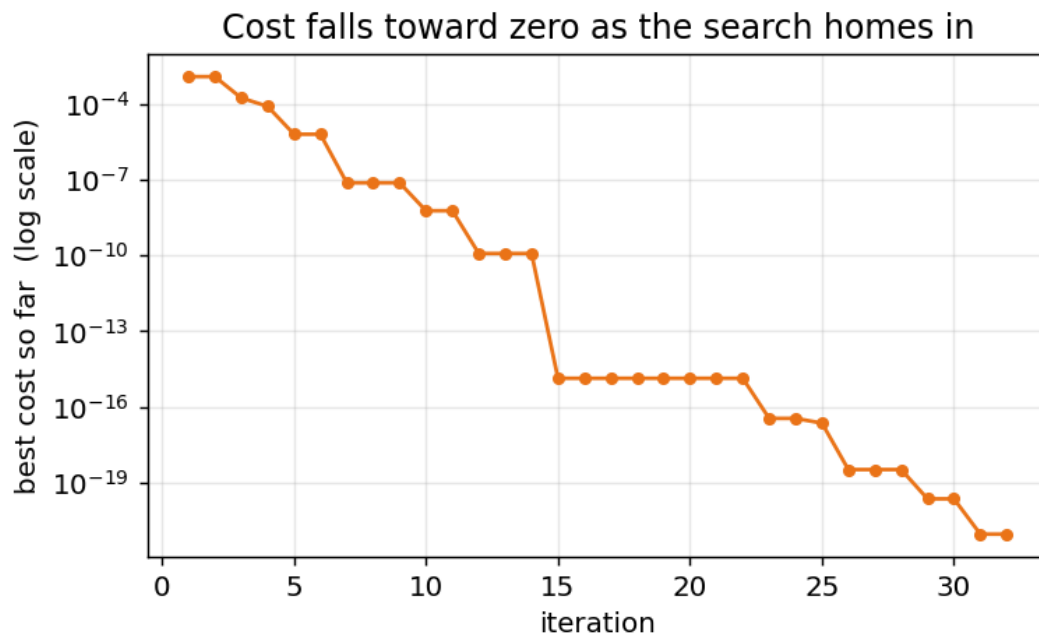
Running it prints:

```
optimize: 1 parameter, analysis 'op', minimizing '(v(out)-0.3)^2'
optimize: converged, objective = 8.99997e-22 after 67 evaluations
      r1 = 2333.33
v(out) = 3.000000e-01
```

The optimizer found $R1 = 2333.33 \text{ }\Omega$ — the exact textbook answer — and left the circuit sitting at that value, so $v(\text{out})$ is now 0.3 V. The objective = 9e-22 is essentially zero (the cost at the bottom of the valley).

Watching it work

Add `-verbose` and the optimizer prints the best cost after each step. Plotting those numbers shows the search closing in on the answer — the cost plunges toward zero:



That is the whole idea in one picture: start somewhere, keep stepping downhill, stop when you can't do better.

4. Example 2 — designing a filter's gain (AC analysis)

Now something a circuit designer actually does. An RC low-pass filter: a resistor R1 into a capacitor C1. Its gain falls with frequency. Suppose we want the gain to be exactly 0.5 at 1 kHz, and we're allowed to choose R1 (with C1 = 100 nF fixed).

We run an AC analysis at just the one frequency of interest (1 kHz) and ask that the magnitude of the output there be 0.5:

RC low-pass: tune R1 so $|gain| = 0.5$ at 1 kHz

```
V1 in 0 ac 1
```

```
R1 in out 1k
```

```
C1 out 0 100n
```

```
.control
```

```
optimize -param R1 1k 100 100k -analysis ac lin 1 1k 1k -minimize
```

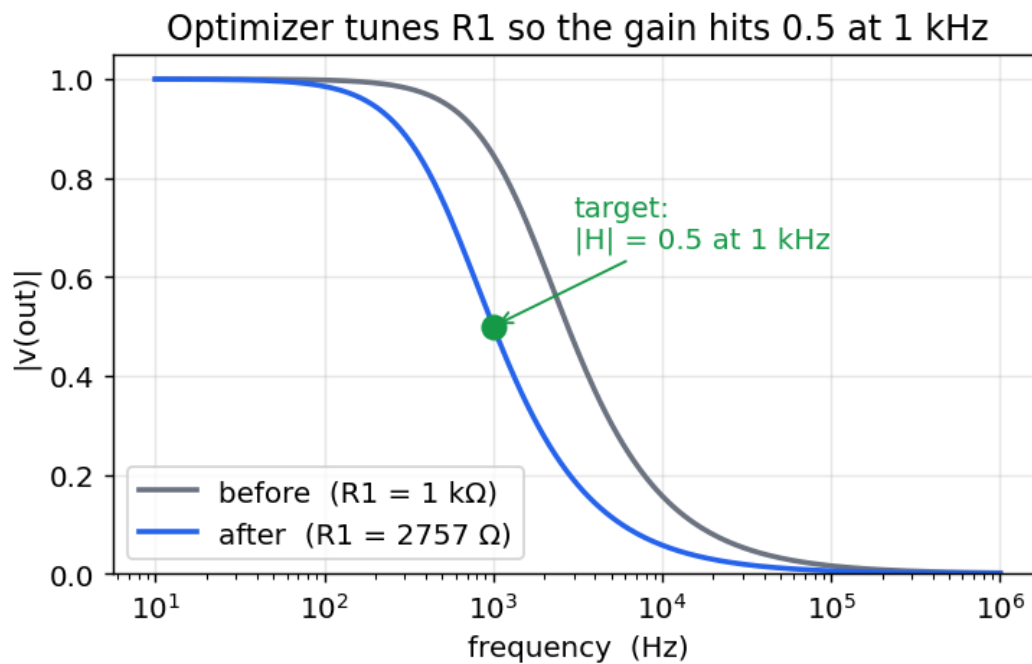
```
↪ (mag(v(out))-0.5)^2
```

```
.endc
```

```
.end
```

`ac lin 1 1k 1k` runs the AC analysis at a single point, 1 kHz. `mag(v(out))` is the gain magnitude there, and we drive $(mag(v(out)) - 0.5)^2$ to zero. The optimizer finds $R1 = 2756.6 \Omega$ (which is the exact value where $1/\sqrt{1+(2\pi fRC)^2} = 0.5$).

Here is the filter's response before (the starting $R1 = 1 \text{ k}\Omega$) and after the optimizer tuned it — the curve slides over until it passes exactly through the target point:



The optimizer only ever looked at the single point at 1 kHz, but because it moved R1 the whole curve shifted with it, landing the 1 kHz gain right on 0.5.

5. Example 3 — two knobs at once

Real designs have several free values. The optimizer handles many parameters together — just add more `-param` flags. Here we tune both resistors of a divider so that two things are true at once:

- the output is 0.4 V, and
- the total resistance is 5 kΩ (so the current drawn from the 1 V source is 0.2 mA).

There is exactly one solution: $R1 = 3 \text{ k}\Omega$, $R2 = 2 \text{ k}\Omega$. We combine the two goals by adding their costs — the total is zero only when *both* are satisfied:

Two-parameter divider design

```
V1 in 0 dc 1
```

```
R1 in out 1k
```

```
R2 out 0 1k
```

```
.control
```

```
optimize -param R1 1k 100 10k -param R2 1k 100 10k
```

```
+ -analysis op
```

```
+ -minimize (v(out)-0.4)^2 + (abs(i(v1))-0.2m)^2
```

```
+ -maxiter 400 -tol 1e-15
```

```
print v(out) i(v1)
```

```
.endc
```

```
.end
```

(Note the + at the start of the continuation lines — that is ngspice's normal way of splitting one long line.) The optimizer reports:

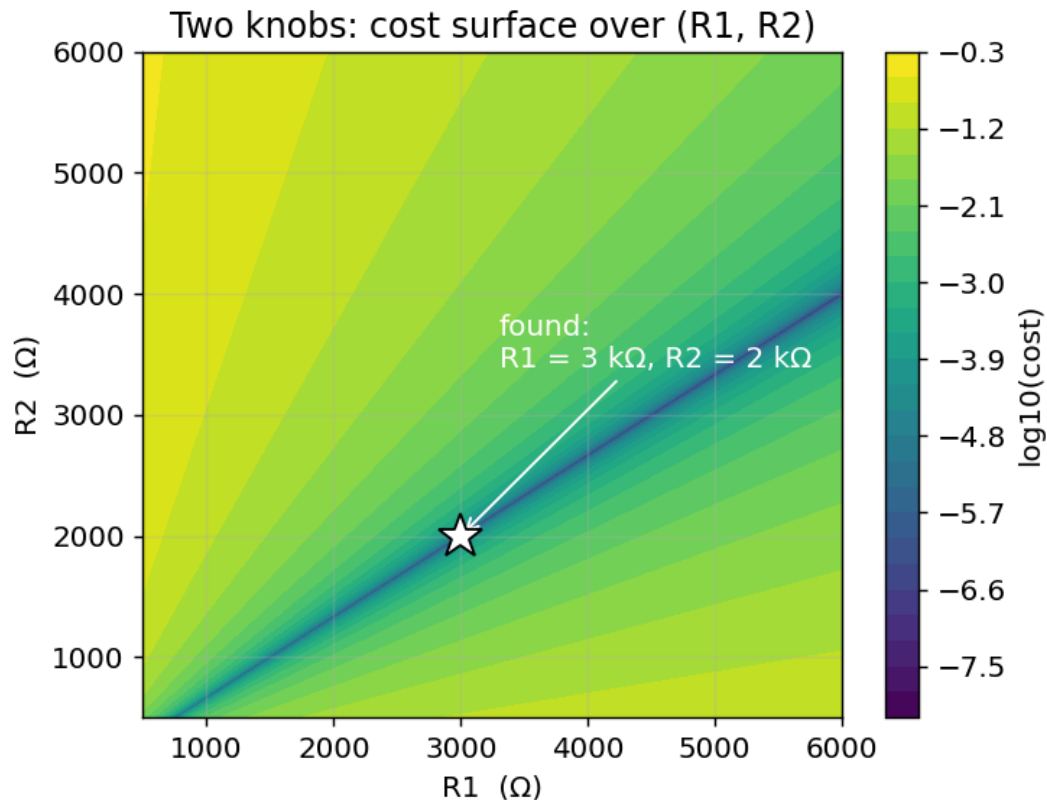
```
optimize: converged, objective = 4e-26 after 192 evaluations
```

```
    r1 = 3000
```

```
    r2 = 2000
```

$v(\text{out}) = 4.000000\text{e-}01$
 $i(v1) = -2.000000\text{e-}04$

Exactly $R1 = 3 \text{ k}\Omega$, $R2 = 2 \text{ k}\Omega$. Now the cost is a surface over two knobs instead of a valley over one. The bright colors are high cost, the dark diagonal streak is the "trench" of good solutions, and the star marks the single point where both goals are met and that the optimizer walked to:



This is why the optimizer is worth having: even with two coupled requirements pulling in different directions, it finds the one combination that satisfies both, from a plain starting guess of $1 \text{ k}\Omega / 1 \text{ k}\Omega$.

6. Example 4 — a time-domain goal (transient analysis)

Goals don't have to be about voltages or gains — they can be about timing. Take an RC circuit charging from a 1 V step. We want the output to reach 0.9 V exactly 1 ms after the step, by choosing $R1$ (with $C1 = 1 \mu\text{F}$).

We run a transient out to 1 ms and ask that the output *at the end of the run* be 0.9 V:

```

RC step: tune R1 so v(out) = 0.9 V at t = 1 ms
V1 in 0 dc 1
R1 in out 1k
C1 out 0 1u ic=0

```

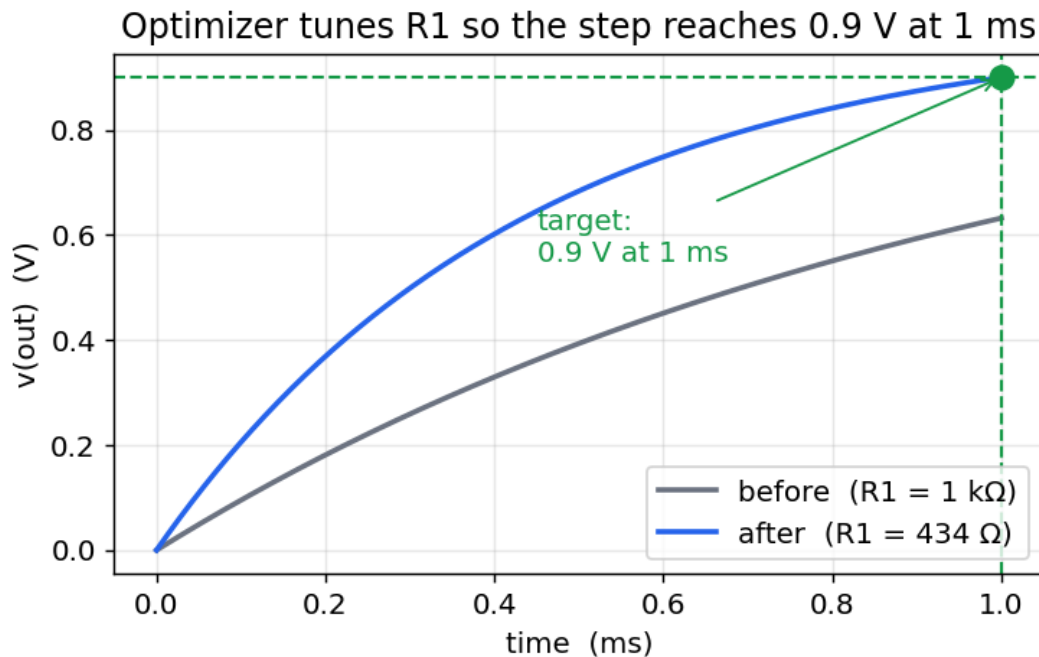
```

.control
optimize -param R1 100 50 5k -analysis tran 2u 1m uic -minimize (v(out)-0.9)^2
.endc
.end

```

Because `tran 2u 1m` stops at 1 ms, the *last* value of $v(\text{out})$ is the voltage at 1 ms, and that is what the cost compares to 0.9. The optimizer finds $R1 = 434.3 \Omega$ (the exact value where $1 - e^{(-1\text{ms}/RC)}$

= 0.9). The step response before and after:



The slow starting curve (1 kΩ) only reaches ~0.63 V by 1 ms; after tuning, the faster curve passes right through the target dot at 0.9 V / 1 ms.

7. Optimizing OSDI / Verilog-A devices

Everything so far used ngspice's built-in components, but the real strength of this build is compiled Verilog-A models (OSDI): you write a device in Verilog-A, compile it to a .osdi file with `openvaf-r`, and load it with `pre_osdi`. The optimizer tunes those devices in exactly the same way — you just point `-param` at an OSDI *instance parameter*.

One rule to remember. For the optimizer to change a Verilog-A parameter, that parameter must be an instance parameter, so mark it with `(*type="instance"*)` in the model. (A plain parameter is a *model* parameter — shared by every instance and not reachable by the per-instance `alter` the optimizer uses.) You then refer to it as `@<instance> [<param>]`.

A Verilog-A resistor

Here is a one-line resistor whose resistance `r` is an instance parameter:

```
// optres.va
`include "disciplines.vams"
module optres(p, n);
    inout p, n;
    electrical p, n;
    (*type="instance"*) parameter real r = 1000.0 from (0:inf);
    analog
        I(p, n) <+ V(p, n) / r;
endmodule
```

Compile it with `openvaf-r optres.va -o optres.osdi` and drop it into the divider from Example 1. The only change is that R1 is now the Verilog-A device N1, and the knob is `@n1[r]`:

OSDI resistor divider: tune the Verilog-A resistor

```
V1 in 0 dc 1
N1 in out rmod
R2 out 0 1k
.model rmod optres r=1k

.control
pre_osdi optres.osdi
optimize -param @n1[r] 1k 100 10k -analysis op -minimize (v(out)-0.3)^2
.endc
.end
```

The optimizer finds $@n1[r] = 2333.33 \, \Omega$ — the same answer as Example 1, now for a compiled device.

Parameter extraction: fitting a diode

The more useful OSDI task is model parameter extraction — you have measured data for a device and want the model parameters that reproduce it. Here is a Verilog-A diode (the Shockley equation) with its saturation current i_s and emission coefficient n as instance parameters:

```
// optdiode.va
`include "disciplines.vams"
module optdiode(a, c);
    inout a, c;
    electrical a, c;
    (*type="instance"*) parameter real is = 1e-14 from (0:inf);
    (*type="instance"*) parameter real n = 1.0 from (0:inf);
    analog
        I(a, c) <+ is * (limexp(V(a, c) / (n * $vt)) - 1.0);
endmodule
```

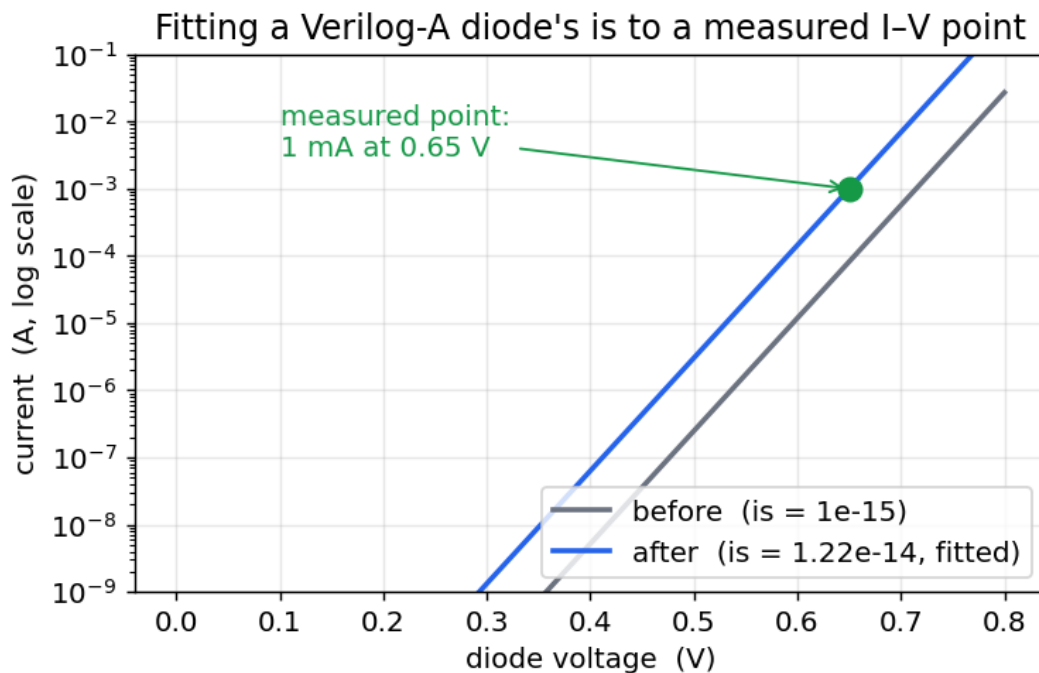
Suppose a measurement says the diode passes 1 mA at 0.65 V, and we want the i_s that reproduces it. We bias the diode at 0.65 V, run an operating point, and drive the current toward 1 mA:

Diode parameter extraction: fit i_s so $I(0.65 \text{ V}) = 1 \text{ mA}$

```
Vd a 0 dc 0.65
N1 a 0 dmod
.model dmod optdiode is=1e-15 n=1

.control
pre_osdi optdiode.osdi
optimize -param @n1[is] 1e-15 1e-16 1e-12 -analysis op -minimize (abs(i(vd))-1m)^2
    ↪ -tol 1e-24
.endc
.end
```

$\text{abs}(i(vd))$ is the current through the diode (what the source V_d must supply). The optimizer reports $@n1[is] = 1.22\text{e-}14$, and the diode now passes exactly 1 mA at 0.65 V. Plotting the whole I-V curve before and after shows the fitted curve sliding onto the measured point:



That is the everyday modeling loop — measure a device, write a Verilog-A model, and let optimize recover the parameters — done entirely inside ngspice. You fit several parameters at once by adding more `-param` flags — and, for several measured points, the dedicated least-squares mode in the next section is the better tool.

8. Fitting to measurements — the least-squares mode

Everything so far minimized one number you wrote by hand (`-minimize <expr>`). But the most common job — *fitting* a circuit or a device model to a set of measurements — is really “make all of these measurements match at once.” You *can* fold them into one expression by hand ($(X-t_1)^2 + (Y-t_2)^2 + \dots$), but that is tedious and throws away useful structure. So `optimize` has a purpose-built mode for it.

Instead of `-minimize`, list each measurement as a **-target**:

`-target <expression> <desired-value> [<weight>]`

The optimizer forms the *residual* $\text{weight} \cdot (\text{expression} - \text{desired-value})$ for each one and drives the sum of their squares to zero — a classic *least-squares* fit. You can give up to 64 targets.

Targets can come from different analyses

Each `-analysis` opens a stage, and every `-target` after it is measured on that stage's results. So a single fit can span several analyses at once — for example a DC operating point *and* an AC response. Here we fit a series R1 and a shunt R2□ C so that the DC gain is 0.4 and the gain at 2 kHz is 0.221:

Least-squares fit across a DC and an AC analysis

```
V1 in 0 dc 1 ac 1
R1 in out 3.3k
R2 out 0 3.3k
C1 out 0 100n
```

```
.control
optimize -param R1 3.3k 500 8k -param R2 3.3k 500 8k
+       -analysis op           -target v(out)      0.4
+       -analysis ac lin 1 2000 2000 -target mag(v(out)) 0.221061
```

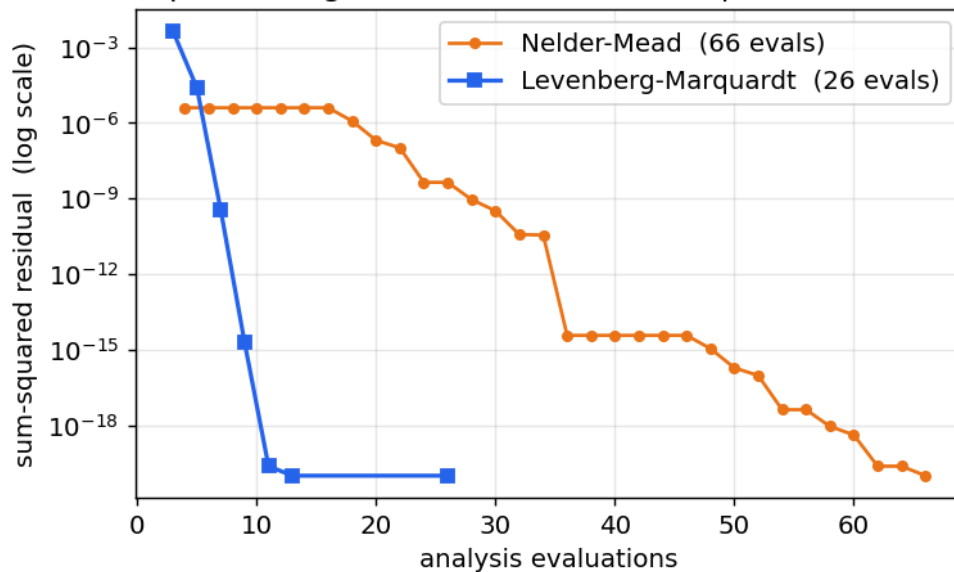
```
.endc
.end
```

The optimizer reports $R1 = 3 \text{ k}\Omega$, $R2 = 2 \text{ k}\Omega$ — the one circuit that satisfies both goals — recovered from a 3.3 k / 3.3 k start.

It fits faster, too

When you write the objective as a list of residuals, the optimizer knows it is a least-squares problem and switches to the Levenberg-Marquardt method — a gradient-based search that estimates the slope of each residual (a *Jacobian*) and steps straight toward the bottom, instead of the slower “shrinking triangle” of Nelder-Mead. On smooth problems this is dramatic: the same two-target filter fit below reaches the optimum in 26 evaluations with Levenberg-Marquardt versus 66 with Nelder-Mead.

Least-squares fit: gradient LM reaches the optimum in fewer runs



You can force either method with **-method nm** (Nelder-Mead) or **-method lm** (Levenberg-Marquardt); by default a **-target** fit uses **lm** and a scalar **-minimize** uses **nm**.

Device parameter extraction, the proper way

The diode fit from the previous section had just one measured point. With two points we can recover both i_s and n at once. Measure the diode’s current at two voltages (here 0.6 V and 0.7 V), then fit both parameters as one least-squares problem — using the optional weight $1/\text{current}$ so each point counts *relatively* even though the two currents differ by more than a decade:

Diode extraction: recover i_s AND n from two I-V points

```
Vd a 0 dc 0.6
```

```
N1 a 0 dmod
```

```
.model dmod optdiode is=5e-15 n=1.0
```

```
.control
```

```
pre_osdi optdiode.osdi
```

```
optimize -param @n1[is] 5e-15 1e-15 5e-14 -param @n1[n] 1.0 0.5 2.0
```

```
+ -analysis dc Vd 0.6 0.7 0.1
```

```
+ -target abs(i(vd))[0] 2.4856e-6 402315
```

```
+ -target abs(i(vd))[1] 6.2326e-5 16045
```

```
.endc
.end
```

`abs(i(vd))[0]` and `[1]` index the two points of the DC sweep. From a deliberately wrong start (`is = 5e-15`, `n = 1.0`) the fit recovers `is = 1e-14`, `n = 1.2` — the exact values the “measurements” came from.

A note on targeting a single point. Like `-minimize`, each `-target` reads the last value of its expression. Use a one-point analysis (e.g. `ac lin 1 f f`) or a vector index (`v(out)[3]`, as above) to pin a specific point.

9. Writing a good cost expression

The cost is the only tricky part, and the recipe is simple: make it zero when the circuit is perfect, and positive otherwise. Some patterns:

Goal	-minimize expression
make X equal a target t	$(X - t)^2$
make X as large as possible	$-X$ (minimizing $-X$ maximizes X)
make X as small as possible	X (or X^2 if X can be negative)
satisfy two goals at once	add them: $(X-t1)^2 + (Y-t2)^2$
weight one goal more	scale it: $(X-t1)^2 + 100*(Y-t2)^2$

X and Y are any ngspice expressions over the results — node voltages `v(out)`, branch currents `i(v1)`, magnitudes `mag(v(out))`, decibels `db(v(out))`, and so on. If the expression produces a whole waveform (as in a transient), the optimizer uses its last value — the value at the end of the run.

10. Tips and common pitfalls

- Give sensible bounds. `lo` and `hi` define the search box; pick a range you know contains a good answer. The starting value `init` should be inside it.
- Different scales are fine. You can optimize a kilohm resistor and a nanofarad capacitor together — the optimizer works in a normalized 0-to-1 version of each range internally, so no parameter dominates just because its numbers are bigger.
- Fast analyses optimize fastest. The optimizer runs your `-analysis` dozens to a few hundred times, so a heavy transient makes the whole thing slow. Use the lightest analysis that captures your goal (an `op` or a one-point `ac` if you can).
- Local minima. Like a hiker in fog, the search can settle into a nearby dip that isn't the deepest valley. If the result looks wrong, try a different starting value or tighter bounds.
- It's silent on purpose. No per-iteration output means the optimization succeeded quietly; add `-verbose` to see the cost each step.
- The circuit is left at the optimum. After `optimize` returns, the winning values are already applied, so `print`, `plot`, `wrdata`, etc. all see the optimized circuit.

11. How it works, briefly

Under the hood, `optimize` uses the Nelder-Mead downhill-simplex method — a classic derivative-free optimizer. For N knobs it keeps N+1 trial points (a “simplex”), and each step reflects the worst point through the others, expanding when that helps and contracting when it doesn't, so the shape tumbles

and shrinks downhill until it settles on the minimum. It needs no derivatives — only the ability to run the circuit and read a number — which is exactly what a SPICE simulation gives it.

For every trial it applies the candidate values with `alter`, runs your analysis, and evaluates your cost expression. It searches in a normalized $[0, 1]$ version of each parameter's range so that very different component scales are treated evenly.

When you give `-targets` instead of `-minimize`, the objective is a sum of squared residuals, and `optimize` switches (by default) to Levenberg–Marquardt: it estimates the slope of each residual with a finite difference (a *Jacobian*), forms the normal equations, and solves a small damped linear system for the next step — decreasing the damping when a step succeeds and increasing it until one does. Exploiting the least-squares structure this way reaches the optimum in far fewer circuit evaluations than the simplex on smooth problems (§8).

The implementation lives in `ngspice-46/src/frontend/com_optimize.c`; the design notes are in [Enhancement-130](#) and [Enhancement-143](#), and a runnable example set is under [examples/optimize_examples/](#).